Will Decker

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Education

2024 – Present
Georgia Institute of Technology – Atlanta, Georgia, USA
PhD, Psychology
Advisor: Prof. Anna (Anya) Ivanova

2020 – 2024
Louisiana State University – Baton Rouge, Louisiana, USA
B.S., Psychology
Advisors: Prof. Christopher R. Cox and Prof. Julie M. Schneider

Manuscripts in-prep

Decker, W., Han, I., Niverthi, M., Raghunath, R., Ivanova, A. A. In-prep. Behavioral evidence for task-specific conceptual representations.

Decker, W., Hari, K., Ivanova, A. A. In-prep. Evaluating large language models as semantic annotators.

Decker, W., Huang, W., Ivanova, A. A. In-prep. Aligning vision-language models with brain responses during story comprehension

Publications

2026
Decker, W., Schneider, J. M. 2026. When reducing noise in developmental EEG data, is EEG best left alone? *Developmental Cognitive Neuroscience*.
In-principle accepted registered report to be published in November 2026.

2025
Fan, T., **Decker, W.**, Momsen, J., Haebig, E., Schneider, J. M. 2025. The facilitatory role of rhyme during implicit and explicit word learning. *Journal of Speech, Language, and Hearing Research.*

2024
Fan, T., **Decker, W.**, Schneider, J. M. 2024. The domain-general neural basis of auditory statistical learning in children. *Neurobiology of Language.*

Selected Conference Presentations

* denotes equal contribution.

2026 **Decker, W.**, Ivanova, A. A. Shared conceptual representations for words and pictures across semantic tasks. Poster presented at the 39th Annual Conference on Human Sentence Processing. Cambridge, Massachusetts, USA.

Decker, W.*, Hari, K.*, Ivanova, A. A. Evaluating the potential of LLMs as semantic task annotators. Poster presented at the 39th Annual Conference on Human Sentence Processing. Cambridge, Massachusetts, USA.

Decker, W.*, Binhuraib, T.*, Ferguson, R., Ivanova, A. A. Modeling the neural basis of reasoning with network logic programming. Poster presented at the virtual 2026 IUCRC BRAIN Interim Meeting.

2024 **Decker, W.** Schneider, J.M., Cox, C. Neural processes of statistical learning. Poster presented at the annual Louisiana State University Discover Day Research Symposium. Baton Rouge, Louisiana, USA.

2023 **Decker, W.**, Schneider J. M., Cox, C. The architecture of auditory statistical learning in the brain: a dynamical functional connectivity study. Poster presented at the Society for the Neurobiology of Language's annual meeting. Marseille, France.

Decker, W., Schneider, J. M. Is EEG really better left alone for developmental datasets. Poster presented at the annual meeting of the Society for Psychophysiological Research (SPR). New Orleans, Louisiana, USA

Decker, W., Fan, T., Haebig, E., Schneider, J. M. Neural mechanisms of novel word learning through rhyme in adults. Poster presented at the Cognitive Neuroscience Society's Annual Meeting. San Francisco, California, USA.

2022 **Decker, W.**, Haebig, E., Schneider, J. M. Anxiety and Word learning in college students. Poster presented at the annual University of Louisiana at Lafayette Undergraduate Research Conference. Lafayette, Louisiana, USA. DOI:https://osf.io/GNM8J

Decker, W., Schneider, J.M. Choosing an EEG system: evaluating the pros and cons and quality of data among two different capping approaches. Poster presented at the annual Louisiana State University Discover Day Research Symposium. Baton Rouge, Louisiana, USA. DOI:https://osf.io/6W4JF

Talks

2026 How similar are you to artificial intelligence?
CEISMC P.E.A.K. Summer Camp, Georgia Institute of Technology

Alignment of vision-language models with brain responses to naturalistic language
School of Psychology, Georgia Institute of Technology

2025 Do words and images have shared conceptual representations?
School of Psychology, Georgia Institute of Technology

2023 Using neuroimaging to examine statistical learning
Student Linguistics Meet-up, Louisiana State University

A pipeline for developmental EEG data
Joint meeting of the Department of Psychology and Department of Communication Sciences and Disorders, Louisiana State University

Funding

2026 Leveraging AI-based predictive modeling to characterize the neural mechanisms of conceptual reasoning.
Agency: AI4Science at the Georgia Institute of Technology

2023 A universal preprocessing-pipeline for event-related potentials (ERPs) in developmental electroencephalography (EEG) research.
Agency: LSU Discover Undergraduate Research Program, Louisiana State University
Amount: $3,000

Honors and awards

2024 Distinguished Undergraduate Researcher Medalist
Louisiana State University

2024 Paul C. Young Award for Outstanding Undergraduate in Psychology
Louisiana State University
Amount: $100

2024 - 2028 President's Fellow
Georgia Institute of Technology
Amount: $22,000

2023 LSU ASPIRE Travel Award
Louisiana State University
Amount: $500

2023 Tiger Athletic Foundation (TAF) Undergraduate Honors Thesis Scholarship
Louisiana State University
Amount: $1,000

2023 LSU Discover Travel Award
Louisiana State University
Amount: $1,500

Mentorship

2026 - Present Devanshi Sood
Undergraduate (Computational Media), Georgia Institute of Technology
Project: *Modeling cortical brain responses with LLM circuits*

2026 - Present Dhriti Raguram
Undergraduate (Neuroscience), Georgia Institute of Technology
Project: *The neural mechanisms of sentence meaning beyond the language network*

2026 - Present Weichen Huang
Undergraduate (Computer science), Georgia Institute of Technology
Project: *Encoding multimodal representations to brain responses of natural language*

2026 - Present Isaac Han
Masters Student (Philosophy), Georgia State University

2026 - Present Saanvi Sundaram
Undergraduate (Neuroscience), Georgia Institute of Technology
Project: *Identifying functional networks involved in sentence plausibility*

2025 - Present Ruthu Raghunath
Undergraduate (Neuroscience), Georgia Institute of Technology

2025 - Present Mala Niverthy
Undergraduate (Neuroscience), Georgia Institute of Technology

Teaching

2024, 2025 Teaching Assistant, Human Language Processing
Instructor: Prof. Anna (Anya) Ivanova

Memberships

2023 - 2024 **Society for Psychophysiological Research**
Student member

2023 - 2024 **Society for the Neurobiology of Language**
Student member

2022 - 2023 **Cognitive Neuroscience Society**
Undergraduate student member

Service

2026 - Present Finance Officer and founding member, Computational Cognition Society (Co-CoSo) @ Georgia Tech

2026 - Present	Supporting member, IUCRC BRAIN Student Network
2025 - Present	Atlanta Swim Association (ASA) Referee (Seasonal)
2025 - Present	Georgia Tech Beekeeping Club